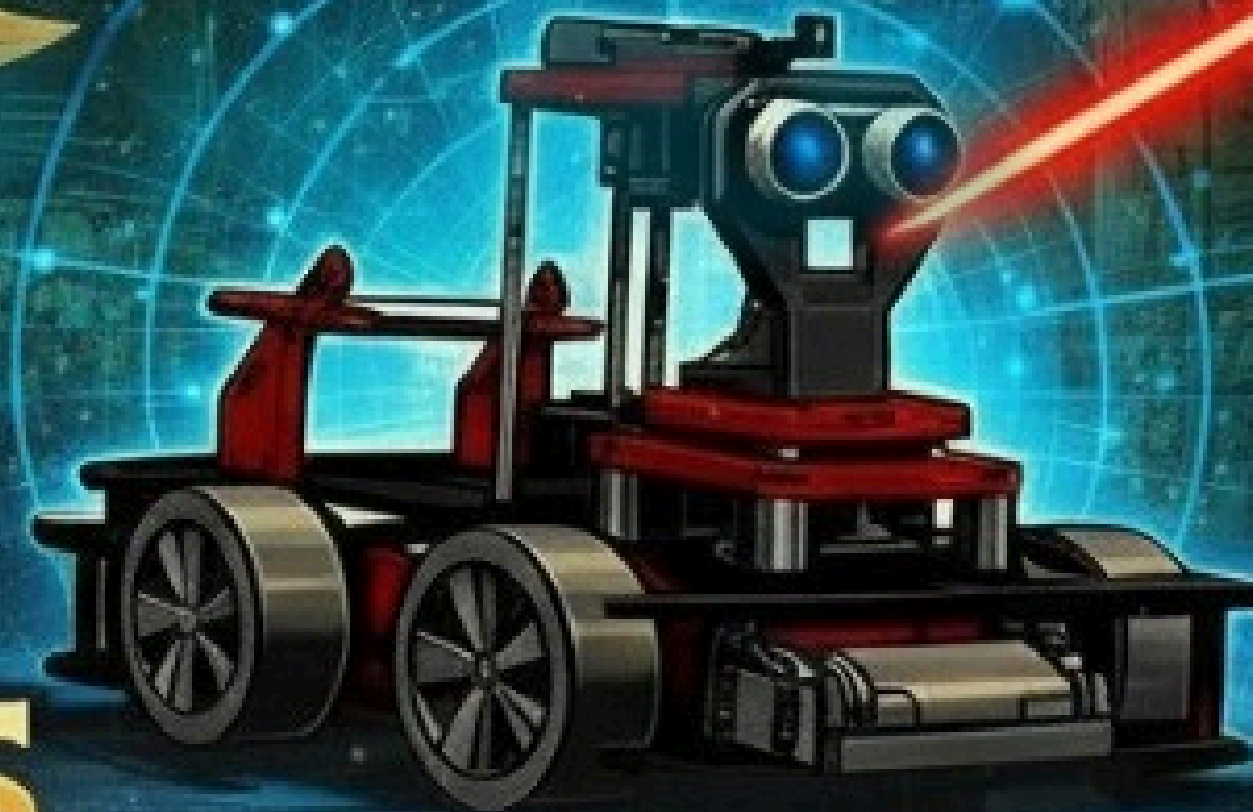


DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING
Graduation project



EYE OF HORUS

Autonomous Ground-Level Ultrasonic Radar & Laser Marking Vehicle

PROJECT TEAM

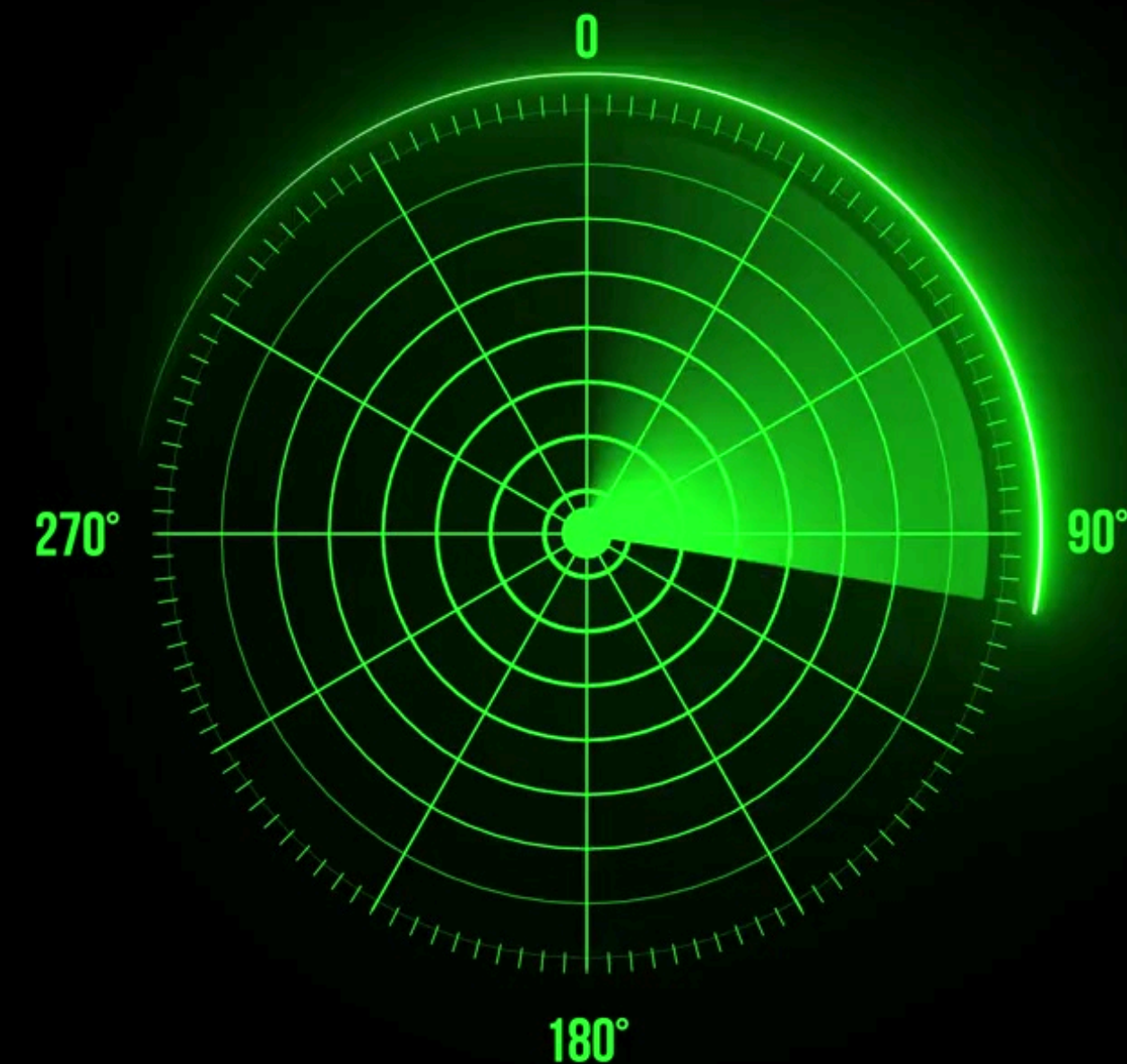
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Ahmed Mustafa
Ahmed Soliman
Abdalla Hossni
Ebru karcı

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- Motivation & Problem Statement
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INTRODUCTION

- Autonomous vehicles can move and operate without direct human control.
- We use sensors to detect their surroundings and make decisions.
- In this project, an autonomous ground vehicle is designed.
- The vehicle uses an ultrasonic sensor to scan the surrounding area.
- It stops, detects objects, and aims a laser toward the detected object.
- The system is simple, low-cost, and effective for short-range detection.



SCANNING AREA

Motivation & Problem Statement

Motivation

DRIVING FACTORS



Low-cost sensing solutions are needed for education and research.

Providing a low-cost alternative to expensive LiDAR systems for educational and hobbyist near-ground sensing applications.



Enhanced Safety

Enabling safer operation in cluttered or indoor environments where blind spots at ground level are critical hazards.



Traditional radar and LiDAR systems are expensive and complex.

Using laser marking to provide instant visual feedback of detection, bridging the gap between sensor data and human observation.

Problem Statement

CHALLENGES ADDRESSED



Ground-level obstacles are difficult to detect reliably.

Reliably detecting small, low-profile obstacles that are often missed by standard bumper sensors or high-mounted cameras.



Accurate distance and angle estimation is required

Achieving accurate angle and distance estimation for object mapping using a single rotating ultrasonic sensor.



The system must work with limited hardware resources.

Implementing real-time autonomous navigation and decision-making logic on embedded hardware with limited processing power.

PROJECT AIM

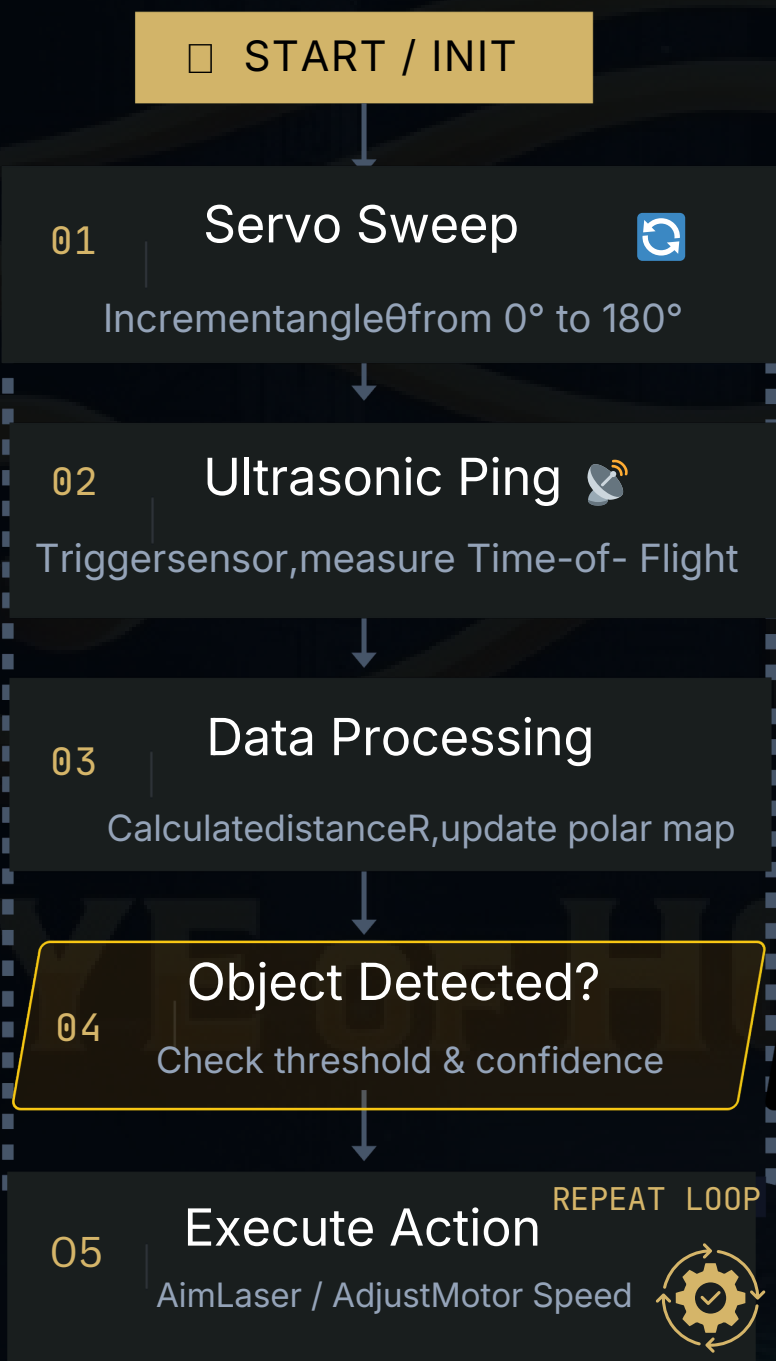
- ◆ **Autonomous Vehicle Design**
Design and construct a compact robotic vehicle chassis integrated with drive systems, power management, and sensor mounts.
- ◆ **Ultrasonic Scanning System**
Implement a servo-controlled ultrasonic sensor mechanism to perform continuous 180° environmental sweeps for wide-area coverage.
- ◆ **Real-time Data Processing**
Develop algorithms to instantly compute object range and bearing (polar coordinates) from raw sensor echoes.
- ◆ **Indicate object location using a laser**
Integrate a laser targeting mechanism that physically marks detected obstacles to provide immediate visual verification of sensor data.
- ◆ **Autonomous Navigation**
Create decision-making logic that identifies free sectors in the scan map and navigates the vehicle without human intervention.



STATUS
LOCKED

PRECISION
High

System Overview



OPERATING LOOP CYCLE

SCAN

>

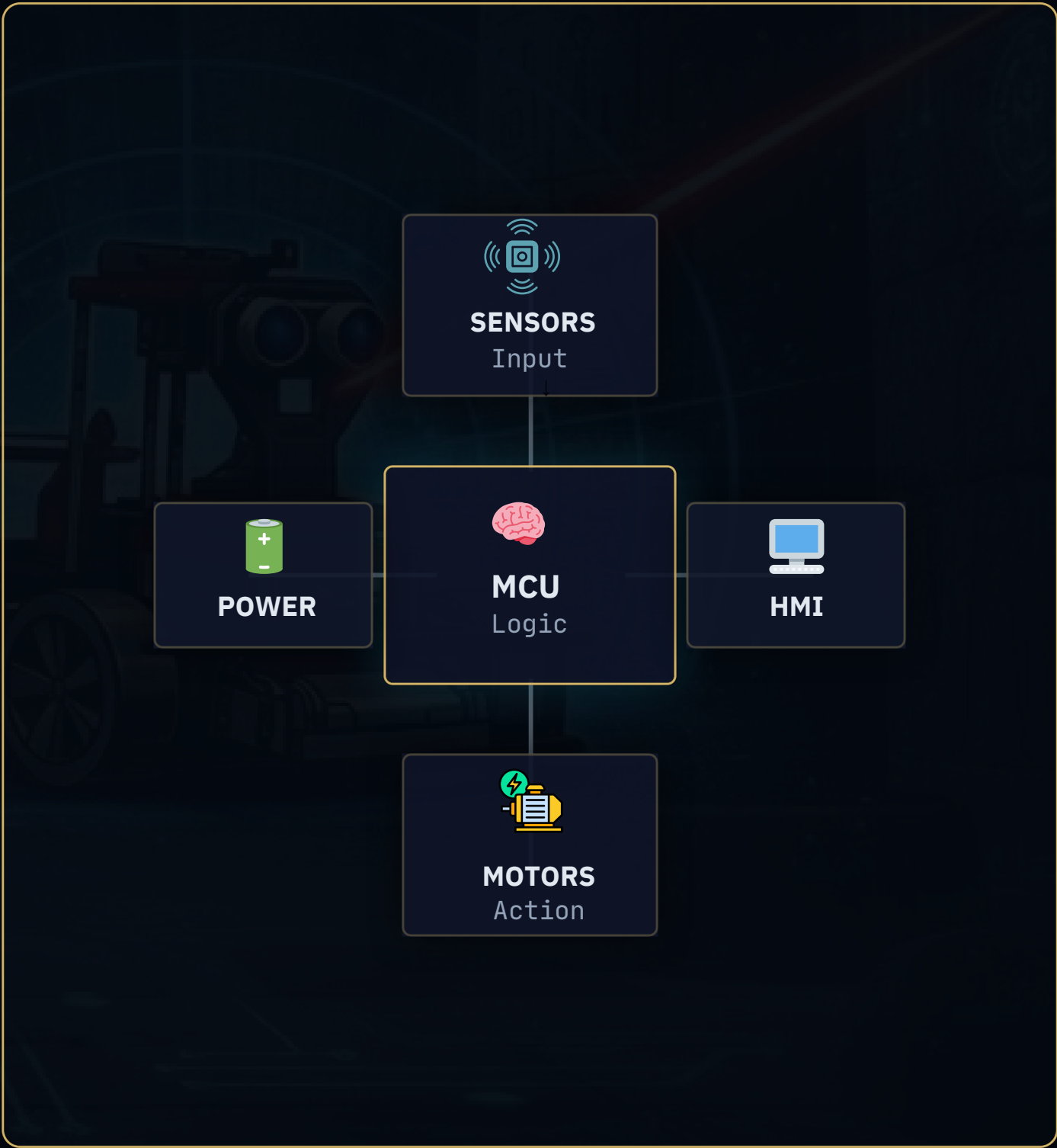
DETECT

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DECIDE

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Action



System Concept & Working Principle

Operational Logic

- 

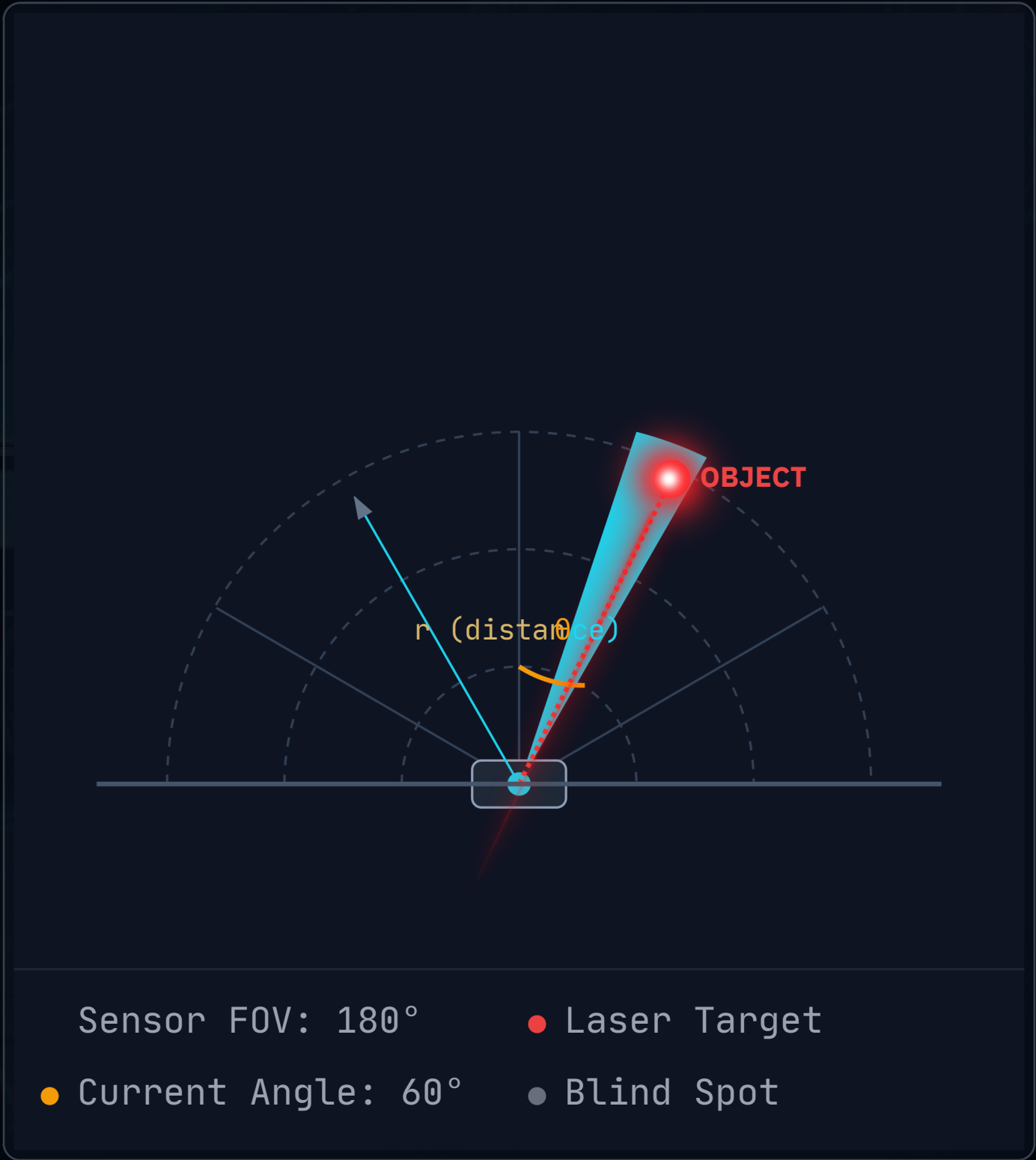
The vehicle moves forward autonomously
The vehicle moves forward automatically without human control using a microcontroller-based control system. It follows a predefined logic to move smoothly and stops at set intervals to scan the environment.
- 

Rotational Scanning
Servo motor sweeps ultrasonic sensor across a 180° field of view (FOV). System continuously updates angular position.
- 

Detected Objects
Detected objects are identified based on distance thresholds, and when an object is within the predefined range, the system recognizes it as a valid target and takes action accordingly
- 

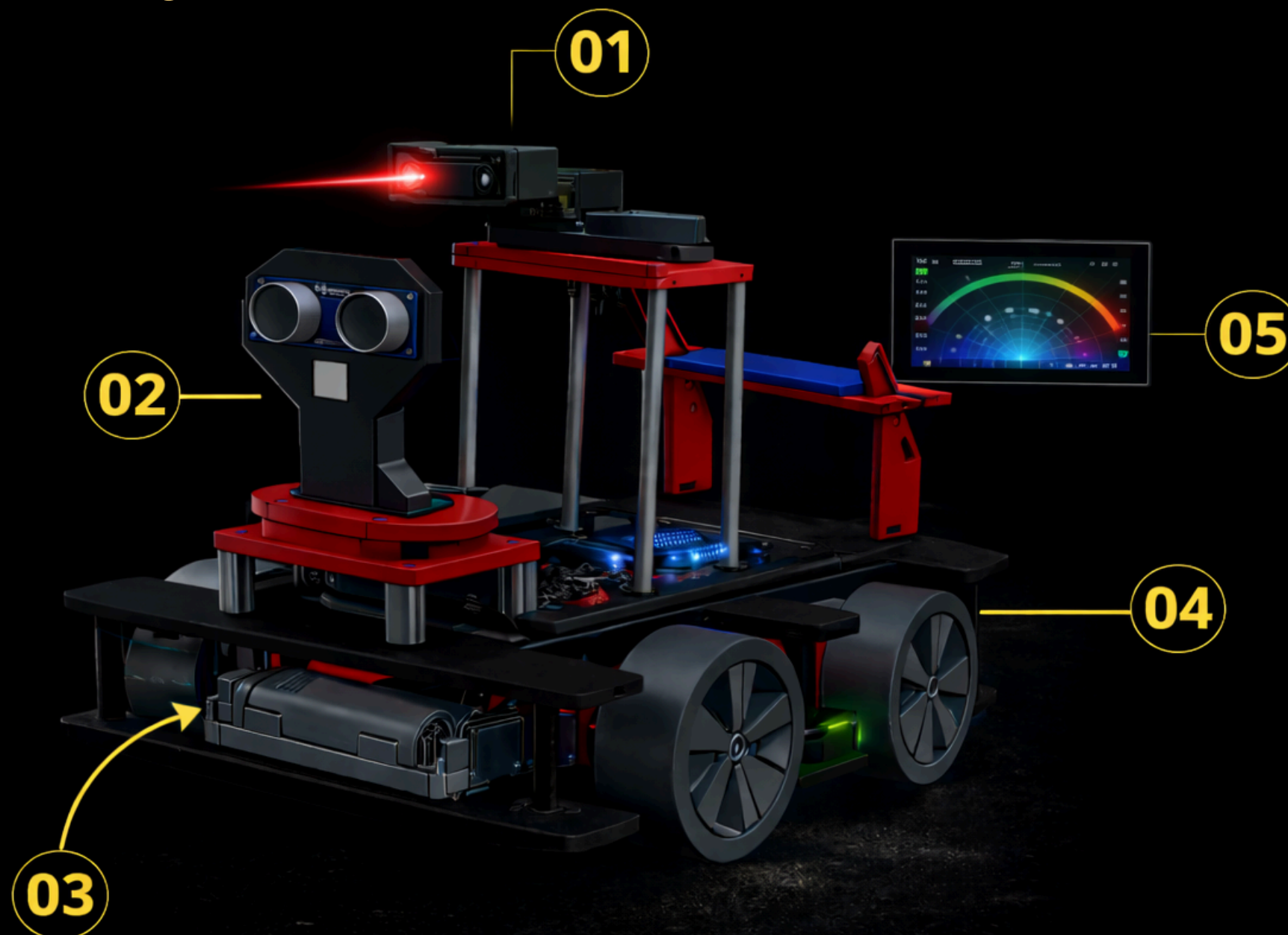
Laser Aiming
The laser aligns with the detected object direction by rotating to the same angle as the ultrasonic sensor, allowing the detected target to be visually marked

SCANNING SUBSYSTEM: ACTIVE



Mechanical & Hardware Design

Car Assembly



01 Laser aiming

Laser module for precise pointing toward detected objects.

02 Ultrasonic Sensor

Measures distance and detects obstacles in front of the vehicle.

03 Drive System

Provides movement and steering using dual DC motors.

04 Power Unit

Supplies power to all system components via a Li-ion battery

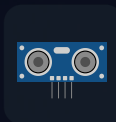
05 Detection Display

Shows real-time object detection and distance data.

Hardware Components

 Sensing & Actuation

INPUT / OUTPUT MODULES



Ultrasonic Module (HC-SR04)

Primary ToF sensor for distance measurement. Capable of 2cm- 400cm range with approx. 3mm accuracy.



Pan Servo Motor

Standard servo providing $\geq 180^\circ$ horizontal sweep for radar-like scanning coverage.



Laser Module & Driver

Class 2/3R laser diode with dedicated driver and safety interlock for visual target designation.



DC Gear Motors

High-torque DC motors with optical encoders for precise vehicle odometry and speed control.



Compute & Power

PROCESSING & ENERGY



Microcontroller (MCU)

Main control unit (e.g., Arduino/STM32) handling sensor timing, motor PWM, and logic execution.



Motor Driver (H-Bridge)

L298N or equivalent dual H-bridge driver for bidirectional control of DC motors.



Power Management

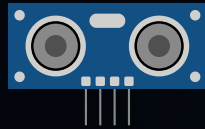
Li-ion battery pack with voltage regulators (5V logic, Vmotor) and overcurrent protection.



HMI Interface

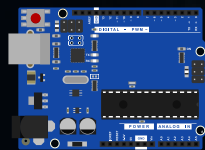
Status LEDs, piezoelectric buzzer for alerts, and emergency stop (E-stop) switch.

Software & Control Logic



Ultrasonic Sensor

Measures the distance to nearby objects using ultrasonic waves.
Provides real-time distance data to the control system



Microcontroller (Arduino Uno)

Acts as the main control unit of the system.
Processes sensor data and controls all connected components.



Servo Motor

Controls angular movement with high precision.
Used to align the laser toward the detected object.



Laser / LED Indicator

Provides a visual indication of the detected object's position.
Helps illustrate the direction of detection.



Buzzer

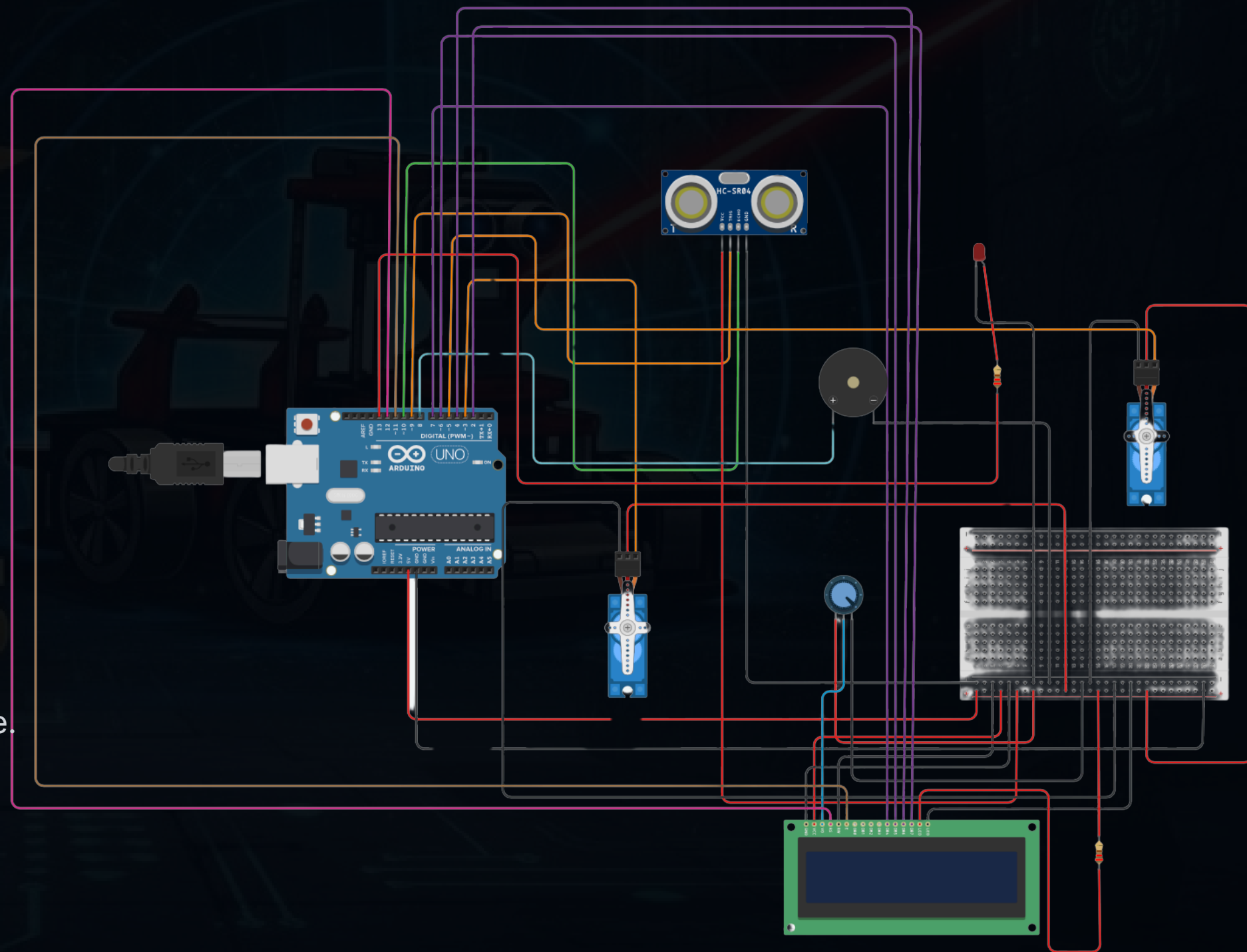
Generates an audible alert when an object is detected within a set distance.
Used for warning and system feedback.



LCD Display

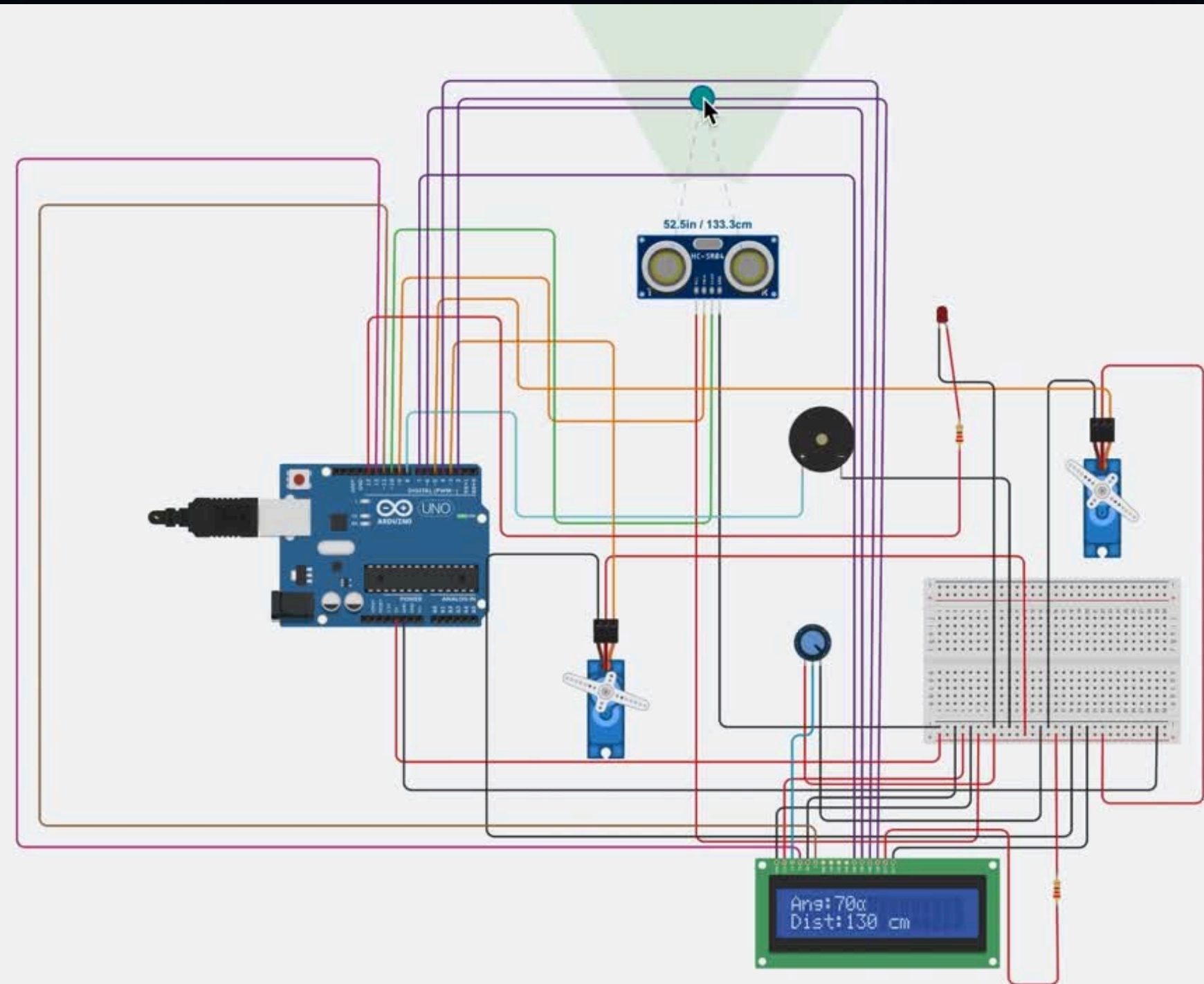
Displays real-time system information such as distance and angle.
Used for monitoring system operation.

KERNEL: ACTIVE



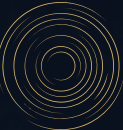
Simulation Demonstration (Software Validation)

- Real-time ultrasonic distance measurement
- Servo-based scanning motion
- Visual indication of detected objects
- Live system response to distance changes



Object Detection Logic

Processing Pipeline

- 

1. Echo Validation

Ultrasonic sensor collects distance data at different angles.
- 

2. Signal Filtering

Invalid and noisy readings are filtered out.
- 

3. Angle Association

Distance data is mapped with the corresponding scan angle.
- 

4. Object Confirmation

Nearby detections are grouped as a single object.
- 

5. Action Trigger

A detection is confirmed only if it persists across multiple scans



Laser Aiming Mechanism

Targeting & Control Logic

- 

Co-Axial Alignment

Laser module is mounted on a servo motor aligned with the ultrasonic sensor for accurate targeting.
- 

Angle Calculation

The detected object angle is used to calculate the servo rotation for laser alignment.
- 

Stable Positioning

The laser is activated only after the servo reaches the correct position.
- 

Safety Control

Laser operation is limited by distance thresholds to ensure safe use.

MARKING SEQUENCE

DETECT

>

ALIGN

>

SAFE CHECK

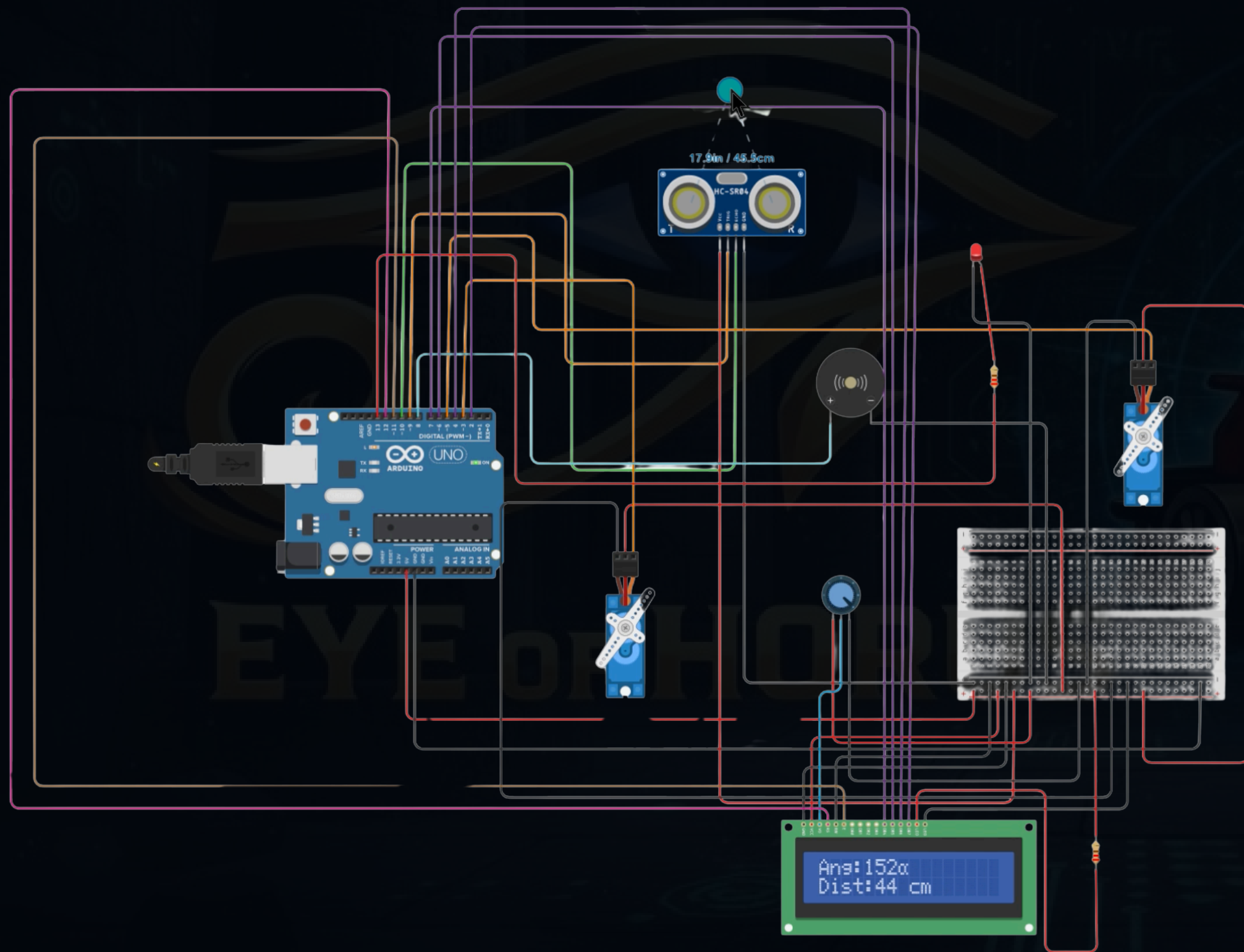
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Aim



Experimental Setup & Results

Indoor Controlled Track



01 Test Environment

System behavior was evaluated through software simulation and design-level analysis.

02 Test Scenarios

Different virtual objects and positions were used to analyze detection logic and system response.

03 Evaluation Criteria

Distance thresholds, detection consistency, and angle alignment were used to evaluate system behavior.

04 Verification Method

Simulation outputs and logical flow were reviewed to verify correct object detection and laser aiming.

System Evaluation

◆ Sensor Behavior Analysis

Ultrasonic sensing performance was analyzed through simulation and design evaluation, considering different object types and detection angles

◆ Noise and False Detection Handling

Basic filtering and thresholding logic were reviewed to reduce unstable readings and false detections in the proposed system

◆ Laser Aiming Verification

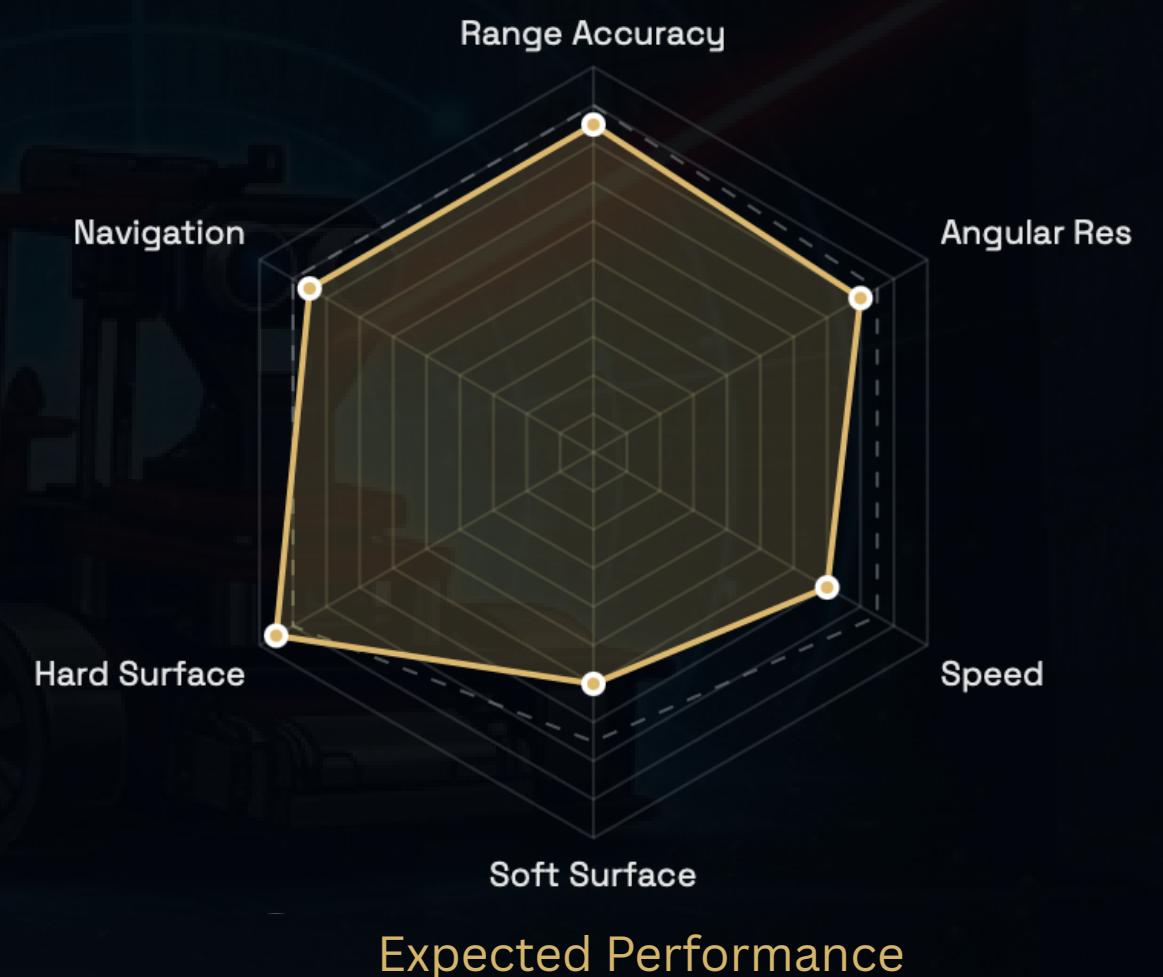
Laser alignment behavior was evaluated conceptually to confirm correct pointing toward the detected object direction

◆ System Limitations

Detection accuracy is expected to decrease for soft surfaces and sharp angles, which is a known limitation of ultrasonic sensing

◆ Objective Validation

The evaluation confirms that the proposed design meets the project objectives and is suitable for future hardware implementation



Applications

◆ Autonomous Navigation

The system can be used in autonomous ground vehicles for basic navigation and obstacle avoidance in controlled environments

◆ Object Detection & Localization

Ultrasonic sensing allows short-range object detection and localization based on distance and angle information.

◆ Embedded Systems Development

The project serves as a practical platform for developing and testing embedded control and sensor integration techniques

◆ Educational Demonstration Platform

The system is suitable for teaching radar-inspired sensing and signal processing concepts in an intuitive and visual way.

◆ Robotics Prototyping

The design can be used as a prototype platform for testing sensor-based perception algorithms in robotics applications.



Future Work



LiDAR-Based Perception Upgrade

Upgrading ultrasonic sensing to LiDAR for higher accuracy, longer range detection, and improved environmental mapping



Advanced UGV Platform Development

Developing the system into a more capable unmanned ground vehicle (UGV) with improved mobility and navigation.



Multi-Sensor Fusion

Integrating LiDAR, cameras, and ultrasonic sensors to increase detection reliability in complex environments.



SLAM-Based Navigation

Implementing SLAM algorithms to enable autonomous mapping and localization in unknown environments.



Wireless Communication & Monitoring

Adding Wi-Fi or cellular communication for real-time monitoring, data transmission, and remote control.



Conclusion

◆ Autonomous Radar-Inspired Vehicle Design

A conceptual design of an autonomous ground vehicle based on radar-inspired sensing using ultrasonic technology was developed.

◆ Scanning and Detection Concept

The system concept demonstrates distance and angle-based object detection using a rotating ultrasonic scanning head.

◆ Visual Target Indication

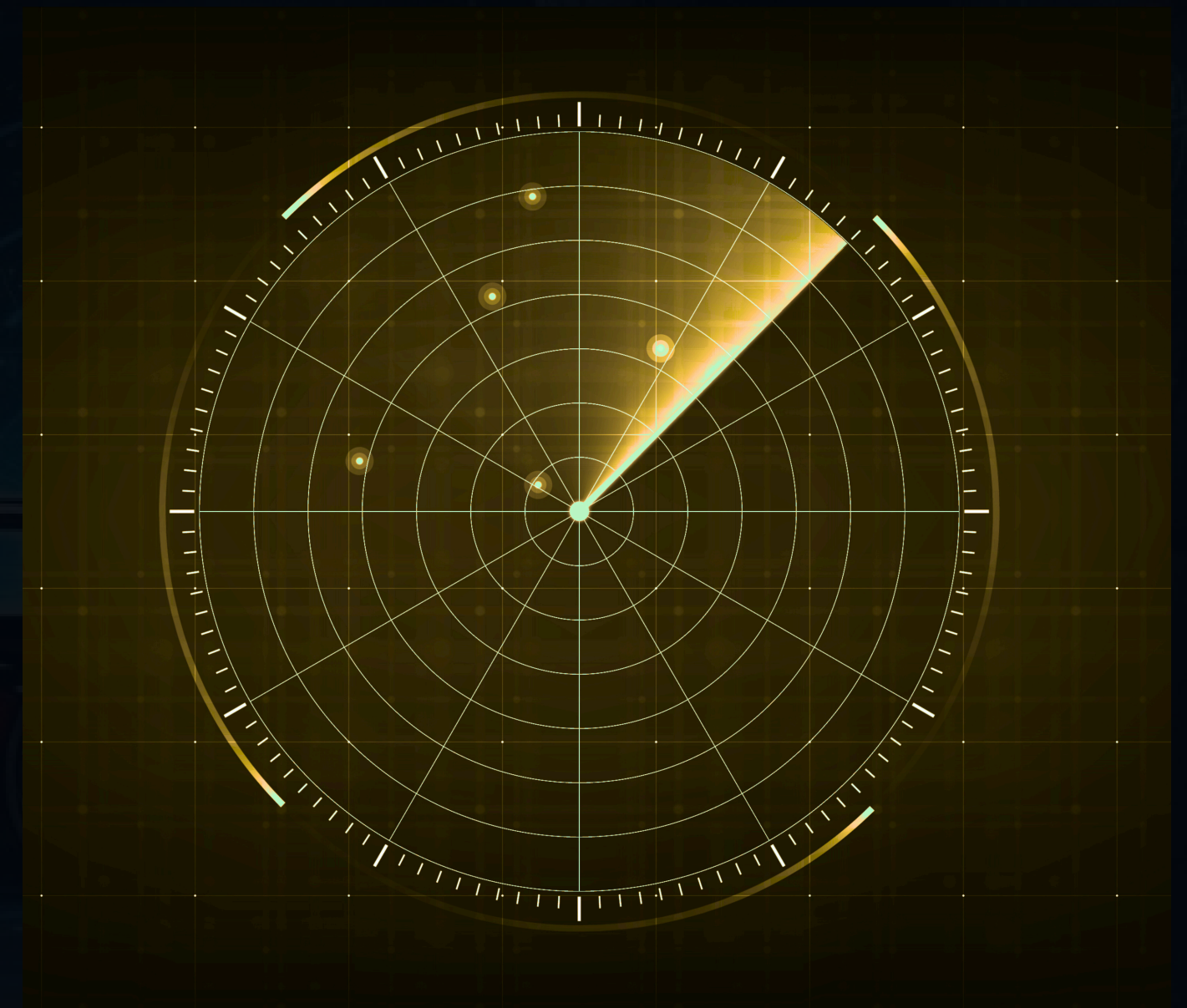
A laser-based visual marking approach was proposed to provide intuitive confirmation of detected objects.

◆ Autonomous Navigation Strategy

An autonomous navigation logic based on free-sector analysis was designed to support obstacle avoidance in controlled environments.

◆ Design Feasibility Validation

System feasibility was evaluated through design analysis and simulation, confirming suitability for educational and research applications



DESIGN OBJECTIVES

DEFINED

SYSTEM STATUS

CONCEPTUAL

IMPLEMENTATION

Next semester

EYE OF HORUS

PROJECT PRESENTATION COMPLETE

THANK YOU

We are now open for questions.

Acknowledgements

We would like to thank our supervisor and jury members for their valuable time and evaluation.

SUPERVISED BY

Prof. [İsmail Hakkı ALTAŞ]

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Prof. / Dr. [V.yurchenko]

Prof. [Hüseyin köse]